

Buffet Working Group

Test Case 2



Version 3
August 4, 2025

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- **Version 3**

- August 4, 2025
- Changed all references to “Test Case 2a” to “Test Case 2” as there will be no 2b
- Updated JAXA data table to indicate Test Case 3 data have been released

- **Version 2**

- April 15, 2025
- Added geometry file names for 1.22 and 2.29 cases

- CRM wing/body/tail0 configuration
- Unsteady CFD with committee-supplied static wing geometry (no FSI)
- Simulations executed at wind-tunnel scale
 - Maximize consistency with dimensional FEMs
 - Geometry and grids are model scale (2.16%)
- **Test Case 2**
 - Released 3/18/25
 - Detailed comparisons to experimental data (Data Set B.1)

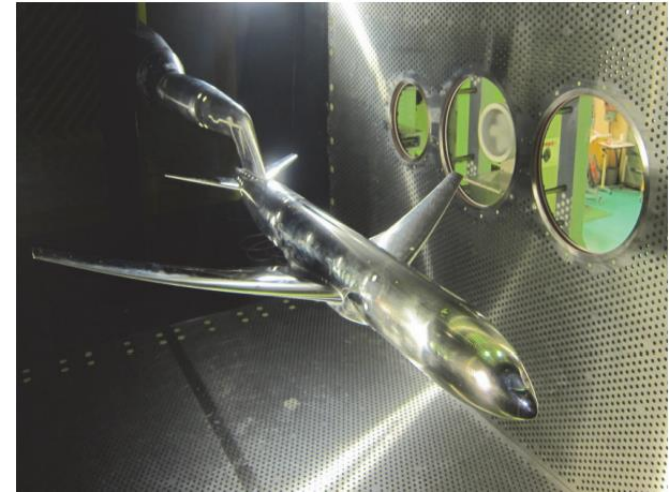
Koike, S., Ueno, M., Nakakita, K., and Hashimoto, A. "Unsteady Pressure Measurement of Transonic Buffet on NASA Common Research Model." AIAA Paper 2016-4044. AIAA Applied Aerodynamics Conference. Washington, DC. June, 2016.

Paper: <https://commonresearchmodel.larc.nasa.gov/wp-content/uploads/sites/7/2018/01/AIAA-2016-4044.pdf>

Data: <https://cfdws.chofu.jaxa.jp/apc/dpw/upc.html>

Experimental Test

- **2.16% scale CRM (80% scale of NASA model) tested in JAXA 2m x 2m transonic wind tunnel**
 - Reynolds numbers of 1.5 and 2.3 million
 - Rich data set of steady and unsteady data
- **Model details**
 - 80% scale NASA CRM (2.16% full-scale vehicle)
 - Wing/body/tail
 - Wind-off wing shape is the as-defined (in 2008) 1-G shape (same as NASA CRM)



Data Set	Test Case	Wing	Re	Alpha	Static, Loaded Deformation	F&M	Static Taps	Kulites	Oil Flow	Wake PIV	TSP	PSP	uPSP	Strain Gauge	FEM	Release Status
A.1	---	Steady	~2.3	-2 to 6 every ~1.2 deg	X	X	X		X	X						Public
A.2	---	Steady	~2.3	-2 to 7	X	X	X				X	X				Requested
B.1	2	Unsteady Wing #1	~1.5	1.22, 2.29, 4.84, 5.89	X	X		X								Public
B.2	3	Unsteady Wing #2	~2.3	-2 to 7		X		X					X	X	X	Public

- **Committee-supplied CAD for experimentally-measured deformations**

- CRM wing/body/tail (0 deg tail deflection)
- Note “updatednose” wording in CAD files
- 4.84 and 5.89 deg available now
- 1.22 and 2.29 deg being developed

<https://cfdws.chofu.jaxa.jp/apc/dpw/geometry.html>

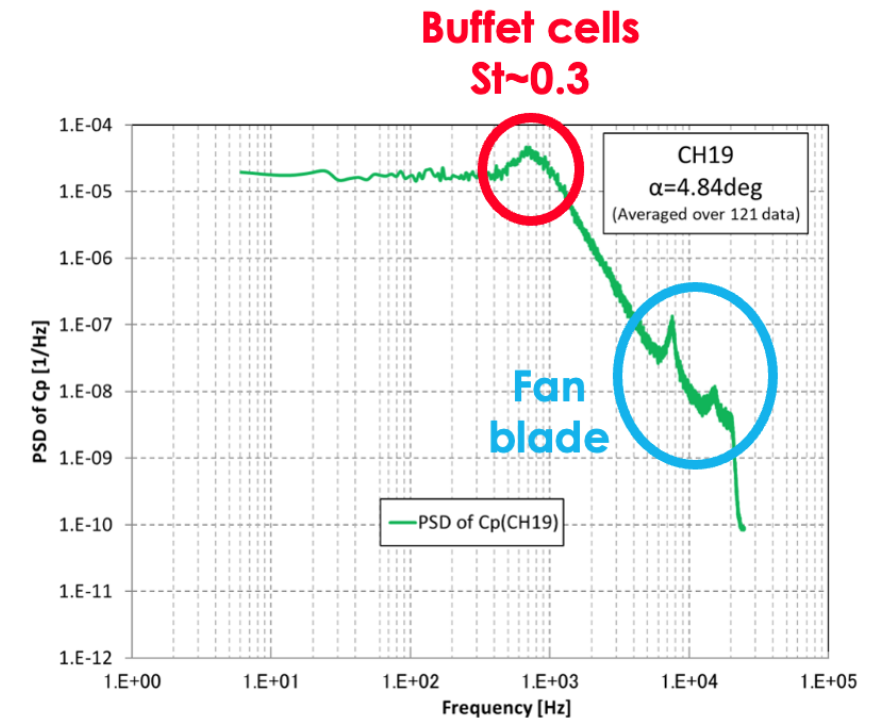
- **Committee-supplied URANS grids**

- Cadence, Helden, and Ames
- Recommended to use model-scale grids
- Model scale maximizes postprocessing consistency and FEM compatibility
- Scale-resolving schemes will need custom grids
- Provide custom grids to the committee for posting to the DPW site

https://dpw.larc.nasa.gov/DPW8/Buffer/Test_Case_2

Alpha	File Name
1.22	wbh_aoa122_buffet_deformed_updatednose.iges.gz
2.29	wbh_aoa229_buffet_deformed_updatednose.iges.gz
4.84	wbh_aoa484_buffet_deformed_updatednose.iges.gz
5.89	wbh_aoa589_buffet_deformed_updatednose.iges.gz

- Recommended to use your best practices from Test Case 1b
- Freestream settings
 - Mach 0.85, $Re_c=1.515m$ (based on chord length), $T_{total} = 323.0\text{ K}$ (121.7 F)
 - Alpha: 1.22, 2.29, 4.84, 5.89
- Experimental conditions (for reference):
 - $P_{total} = 80\text{ kPa}$
 - Trip dots at 10% chord on wing
 - Investigating location on nose and tail (an update will be provided in the future)
- Grids
 - Baseline grid is Medium (L3)
 - Grid convergence study optional for $AoA=4.84$
- Optional sensitivities
 - Time step, simulation length, turbulence model, etc.



- **Goal**
 - Resolve frequency peak at $St=0.3$
 - Capture as much of the spectra as reasonably possible
- **Recommended baseline settings**
 - 30 CTU after initial solution stabilized
 - 100 time steps per CTU
 - More CTU may be required to resolve frequency at high resolution
- **Limitations**
 - Computational resources will limit the user's selected time step and simulation time
 - Utilize your best practice for iterations per CTU and simulation length

- **Follow these instructions**

<https://aiaa-dpw.larc.nasa.gov/postprocessing.html>

- **Required data**

- Forces and Moments

DPW8-AePW4_UnsteadyForceMoment_v5.dat

- Surface cuts

DPW8-AePW4_UnsteadySectionalCuts_v5.dat

- Spectral content

DPW8-AePW4_UnsteadySpectra_v5.dat

- **Spectral content at four locations**

- Section E ($\eta = 0.5$): $x/c = 0.3040$ and 0.7903

- Section F ($\eta = 0.6$): $x/c = 0.3097$ and 0.7677



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